

Frank Shan

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HIGHLIGHTS

- **Full-stack** developer specialized in translating **business requirements into production-ready** systems. Shipped customer-facing software quickly and reliably across backend, automation, and AI-assisted workflows.

EDUCATION

University of Waterloo | *CGPA: 92.3/100* Sep 2024 - Apr 2029
Bachelor of Computer Science, Honours Computer Science, AI Specialization

EXPERIENCE

Software Developer Intern | *Richmond, BC* Sep 2025 - Apr 2026
Ziyutec Inc.

- **Owned** the end-to-end planning and development of a [global B2B marketplace](#) for a startup initiative, **leading** system design, implementation, and deployment (AWS) across the **entire SDLC**.
- Architected a Role-Based Access Control framework (NestJS, PostgreSQL) to support extensible user roles; balanced database **normalization** against **query performance** via denormalization and data **integrity** checks.
- Engineered a **caching** layer to decouple client-side state from **atomic** database transactions, reducing **user friction** while ensuring transactional integrity.
- Built an automation agent for engineering workflow, reducing operational workload by **20%** across a **50-person** engineering team.

Research Assistant | *Waterloo, ON* Feb 2026 - Present
University of Waterloo

- Designed comparative experiments for a probabilistic programming language PAL that compiles structural rules into tensor operations. Built a **hybrid neuro-symbolic** framework for training Named Entity Recognition models to enforce strict linguistic constraints and improve early-epoch model convergence.
- Structured reproducible experiments and analysis pipelines for model comparisons, enabling faster iteration on constraint-driven NER training strategies.

PROJECTS

Resume Generator v1 | *Python, Jinja2, prompt_toolkit, LaTeX, LangGraph* [github](#) 

- Built a modular pipeline that transforms structured YAML resume data into Jinja2-rendered LaTeX artifacts and isolated generation outputs.
- Designed a keyboard-only CLI for resume generation, configuration, and human-in-the-loop review using prompt_toolkit.
- Added an AI-ready selection stage so resume content can be matched to job descriptions before rendering and compilation.

Marketplace Platform | *NestJS, PostgreSQL, AWS*

- Led architecture and implementation of a production marketplace product from blank repo to deployment.
- Designed backend workflows that balanced product speed with transactional correctness.

SKILLS

Programming Languages: `built-in method items of dict object at 0x115b3b740`

AI and ML: `built-in method items of dict object at 0x115b3b6c0`

Frameworks: `built-in method items of dict object at 0x115b3b9c0`

Platforms: `built-in method items of dict object at 0x115b47b00`

PROFESSIONAL CERTIFICATES

Stanford Online December 2025
XCS224N Natural Language Understanding [credential](#) 